

system an available IP address from the DHCP server in your network. Note that it may not be the same IP address that your machine fetches each time you issue *dhcpclient*.

Interface cards

Machines are networked either via network cables or using wireless protocols. LAN cards used for networking are known as Ethernet and wireless LAN (WLAN) cards. We interface the network via this outlet. In *nix platforms, Ethernet cards or WLAN cards are denoted as *eth0*, *eth1*, etc, or *wlan0*, *wlan1*, etc, respectively.

ifconfig

We have *ifconfig*, a.k.a the interface config, for setting up a network on the machine. To get information about the availability of interface devices available on the current machine, open a terminal and execute the following as the root:

```
[slynnux@grubbox ~]$ ifconfig -a
eth0    Link encap:Ethernet HWaddr 00:1C:23:FB:37:22
        inet6 addr: fe80::21c:23ff:fe3b:3722/64 Scope:Link
        UP BROADCAST MULTICAST MTU:1500 Metric:1
        RX packets:9724 errors:0 dropped:0 overruns:0 frame:0
        TX packets:2720 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:2400589 (2.2 MiB) TX bytes:645396 (630.2 KiB)
        Interrupt:17

lo       Link encap:Local Loopback
        inet addr:127.0.0.1 Mask:255.0.0.0
        inet6 addr: ::1/128 Scope:Host
        UP LOOPBACK RUNNING MTU:16436 Metric:1
        RX packets:76320 errors:0 dropped:0 overruns:0 frame:0
        TX packets:76320 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:0
        RX bytes:87151068 (83.1 MiB) TX bytes:87151068 (83.1 MiB)

wlan0    Link encap:Ethernet HWaddr 00:1C:BF:87:25:D2
        inet addr:192.168.1.143 Bcast:192.168.1.255 Mask:255.255.255.0
        inet6 addr: fe80::21c:bfff:fe87:25d2/64 Scope:Link
        UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
        RX packets:45302 errors:0 dropped:0 overruns:0 carrier:0
        TX packets:37510 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:31091293 (29.6 MiB) TX bytes:9734025 (9.2 MiB)
```

Here I have three interfaces -- *eth0*, *lo* and *wlan0*, where:

- *eth0* corresponds to the Ethernet card
- *lo* corresponds to a loopback device that points to the localhost network
- *wlan0* corresponds to the wireless LAN card

Static IP addressing

For static IP addressing, issue the following command as the root:

```
#ifconfig <device name> <ip address>
```

For example:

```
#ifconfig eth0 192.168.0.2
```

ifconfig -a gives you details of all interface devices and configurations. In order to receive details of only one Ethernet device, execute *ifconfig eth0*. The following is an example output:

```
[root@grubbox slynux]# ifconfig eth0
eth0    Link encap:Ethernet HWaddr 00:1C:23:FB:37:22
        inet addr:192.168.0.2 Bcast:192.168.0.255 Mask:255.255.255.0
        inet6 addr: fe80::21c:23ff:fe3b:3722/64 Scope:Link
        UP BROADCAST MULTICAST MTU:1500 Metric:1
        RX packets:9724 errors:0 dropped:0 overruns:0 frame:0
        TX packets:2720 errors:0 dropped:0 overruns:0 carrier:0
        collisions:0 txqueuelen:1000
        RX bytes:2400589 (2.2 MiB) TX bytes:645396 (630.2 KiB)
        Interrupt:17
```

Now, let us go about setting the subnet mask. This is done easily with *ifconfig*, as follows:

```
#ifconfig eth0 192.168.0.2 netmask 255.255.255.0
```

Now that the IP address and subnet mask is configured (or reconfigured), how do you get your Ethernet up (i.e., available) and down? The *ifup* and *ifdown* commands help you with that as follows:

```
#ifup eth0
#ifdown eth0
```

Wireless networking

In order to hack a wireless card, we have another utility called *iwconfig*. It works similar to *ifconfig*, but it has lots of additional features that are bonded to wireless cards. If we are using a wireless network with static IP, we can attach our wireless card interface to a network as follows:

```
iwconfig wlan0 essid slynux
...or:
iwconfig wlan0 essid slynux key 8c140b2037
```

...where 'slynux' is the ESSID (that is, the name of the wireless network) and '8c140b2037' is the security key. Of course, you need to replace these variables with the values that hold good in your network. You can also scan and check the availability of wireless network(s) in your vicinity using the *iwlist* command as follows: